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ICE 3.1



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8/8

1. (2 marks) In symmetric key cryptography, what do the terms diffusion and confusion refer to?

- Diffusion refers to fact that each bit in plaintext influences all bits in the ciphertext, whereas confusion ensures that the structure of the key cannot be readily obtained from a close look at the ciphertext. (correct)
- Confusion refers to fact that each bit in plaintext influences all bits in the ciphertext, whereas diffusion ensures that the structure of the key cannot be readily obtained from a close look at the ciphertext
- Both confusion and diffusion refer to the same things
- None of the above

2. (2 marks) Using an ideal substitution cipher with 4-bit input and 4-bit output, (a) what is the size of the table that maps input to output, (b) what is the potential number of possible keys?

- (a) table size 4, (b) number of possible keys 2^4
- (a) table size 16, (b) number of possible keys 2^{16}
- (a) table size 2^{16} , (b) number of possible keys $(2^{16})!$
- (a) table size 16, (b) number of possible keys 16! (correct)

3. (2 mark) Consider DES (or Fiestal cryptosystem). Let cipher $c = E(K, p)$, where p is plaintext and K is the key. Now if bit 1 in plaintext p is changed what is the probability that bit k of the resulting ciphertext c is changed?

- 1
- 1/2 (correct)
- 1/64
- None of the above

4. (2 mark) Reconsider DES (or Fiestal cryptosystem). Let $c = E(K, p)$. Now if bit 1 in plaintext p is changed then how many bits of the resulting ciphertext are LIKELY to change?

- 1
- 32 (correct)
- 64
- Cannot say
- None of the above



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Your work

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